

CHARLESTON EXECUTIVE, SC, USA

WMO: 720606

Lat: 32.701N

Lon: 80.003W

Elev: 5

StdP: 101.26

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00198

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.3	0.1	-12.1	1.3	2.0	-7.9	1.9	4.8	9.9	12.2	8.7	12.2	2.1	0	0.402

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	34.0	26.4	32.9	26.6	32.3	26.5	29.1	31.0	28.0	30.6	27.4	30.1	4.0	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.9	25.5	29.7	27.5	23.4	29.0	26.4	21.9	29.0	95.4	30.4	90.2	29.1	86.3	30.3	34.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.6	7.7	7.0	-3.7	36.7	1.6	1.4	-4.8	37.7	-5.7	38.5	-6.6	39.3	-7.8	40.3	
			DB												
			WB	-5.4	29.0	1.8	0.6	-6.7	29.4	-7.8	29.7	-8.8	30.0	-10.2	30.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.6	9.8	12.0	14.6	19.0	23.4	26.9	28.3	27.6	25.5	20.8	14.6	12.6
	DBStd	7.42	5.28	4.80	4.70	3.54	2.85	2.18	1.69	1.57	2.12	3.68	4.40	4.82
	HDD10.0	153	69	32	14	0	0	0	0	0	0	8	29	
	HDD18.3	949	265	181	133	34	3	0	0	0	0	20	129	184
	CDD10.0	3671	64	88	156	270	417	507	566	546	466	334	146	112
	CDD18.3	1424	2	4	16	54	161	257	308	287	217	95	16	8
	CDH23.3	13444	3	12	71	270	1160	2717	3598	3081	1858	613	49	11
	CDH26.7	4732	0	0	10	32	283	1031	1514	1166	579	116	2	0
Wind	WSAvg	2.9	2.8	3.2	3.5	3.6	3.3	2.8	2.9	2.4	2.6	2.7	2.8	2.6
Precipitation	PrecAvg	1230	87	83	92	76	73	134	145	161	128	104	69	78
	PrecMax	1581	203	268	166	154	164	256	225	283	268	316	185	235
	PrecMin	823	12	30	11	23	14	67	49	89	11	0	2	20
	PrecStd	187	52	53	41	39	43	55	48	56	62	78	48	42
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.7	25.0	27.7	29.1	33.7	35.9	36.0	35.0	32.9	31.1	26.4	24.9
		MCWB	19.1	19.7	19.1	21.1	23.5	25.8	27.2	26.8	25.7	24.9	21.4	21.0
	2%	DB	22.0	22.8	25.2	27.4	31.2	33.8	34.0	33.0	32.1	28.9	24.8	22.8
		MCWB	17.9	18.1	18.7	21.1	23.6	26.0	26.9	26.9	25.7	24.2	20.3	20.2
	5%	DB	19.1	21.1	22.9	26.3	29.0	32.5	32.9	32.4	31.0	27.8	23.0	22.1
		MCWB	16.5	17.3	17.5	20.4	23.2	26.2	27.1	26.9	25.2	23.1	19.7	19.8
	10%	DB	17.7	19.0	21.4	24.9	27.8	31.3	32.3	31.3	29.9	27.1	22.2	20.1
		MCWB	15.5	16.5	17.2	19.8	22.7	25.6	27.0	26.6	25.2	22.5	19.1	18.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.8	21.2	21.3	24.0	25.8	28.6	31.4	31.4	27.5	26.7	23.4	22.7
		MCDB	22.0	23.3	24.5	26.5	30.4	31.6	32.0	31.7	29.6	28.0	25.0	24.1
	2%	WB	18.8	19.5	20.1	23.0	24.9	27.9	29.8	29.1	27.0	25.7	22.1	21.4
		MCDB	20.7	21.8	23.5	25.6	28.5	30.7	31.0	30.2	29.3	27.7	23.6	22.2
	5%	WB	17.6	18.3	19.1	22.0	24.3	27.3	28.2	27.8	26.3	24.7	20.9	20.0
		MCDB	18.4	19.8	21.9	24.8	27.8	30.1	30.9	30.0	29.0	27.1	22.4	20.9
	10%	WB	16.4	17.1	18.2	21.1	23.7	26.6	27.5	27.1	25.7	23.7	19.5	18.6
		MCDB	17.6	18.4	20.2	23.8	26.9	29.8	30.4	29.7	28.5	26.0	21.6	20.1
Mean Daily Temperature Range	5% DB	MDBR	10.1	10.3	10.2	9.6	8.4	8.2	7.6	7.3	8.0	9.7	10.8	9.1
		MCDBR	11.2	11.0	11.1	10.0	9.7	9.6	8.6	8.7	9.3	9.6	10.5	9.8
		MCWBR	7.8	7.3	6.2	5.2	4.4	4.1	4.2	3.9	4.0	5.1	6.8	6.8
	5% WB	MCDBR	9.4	9.5	9.6	8.1	7.8	8.3	7.7	7.8	7.3	7.7	7.6	8.2
		MCWBR	8.1	7.6	6.1	4.7	3.9	4.2	4.7	4.4	3.8	4.8	6.0	7.0
Clear-Sky Solar Irradiance	taub	0.329	0.337	0.366	0.408	0.464	0.496	0.529	0.511	0.435	0.390	0.351	0.335	
	taud	2.540	2.515	2.452	2.317	2.208	2.169	2.089	2.138	2.344	2.449	2.513	2.550	
	Ebn at Noon	892	920	913	882	832	803	774	782	832	848	858	864	
	Edh at Noon	84	94	108	129	145	150	162	152	119	100	85	79	
All-Sky Solar Radiation	RadAvg	2.86	3.53	4.67	5.98	6.46	6.53	6.27	5.66	4.80	4.15	3.19	2.54	
	RadStd	0.17	0.36	0.43	0.40	0.42	0.45	0.38	0.36	0.40	0.44	0.22	0.20	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)	
(47)	Regional (40 neighbors)	+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+200	+139	(47)	

Nomenclature: See separate page